When does dispatch fidelity matter for battery sizing? A regime-classification diagnostic

Julian Quick
Lawrence Berkeley National Laboratory

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Abstract

Tesla Autobidder, Fluence Mosaic, and Wärtsilä GEMS run battery dispatch on machine-learning-driven stochastic optimization. NREL REopt, DTU `hydesign`, and most academic hybrid-power-plant sizing tools embed deterministic LP inner dispatch. Should sizing tools change architecture? We propose a practitioner-runnable regime-classification diagnostic — the $b^{\epsilon}_{\mathrm{sat}}$ overlap test — that any operator can compute on one year of price data to answer the question for their own market. The diagnostic returns "invariance survives" on synthetic AR(1), all three years of Danish DK1 (Energinet, including the 2022 European energy crisis with max €871/MWh), and all three years of ERCOT North Hub (gridstatus, including February 2021 Storm Uri with sustained \$9000/MWh price-cap hours). Despite stochastic dispatch yielding 0.9–37% realized-NPV uplift at the optimum, the optimal battery energy capacity b^*_E is invariant across deterministic vs. multi-lag-ensemble dispatch in every (market, year, cost-form) configuration tested at $\lambda = 0$ (the pure-merchant limit). We sketch a theoretical proposition motivating the diagnostic, then test its prediction empirically: it predicted that multi-day spectral content should break invariance; the data falsify the prediction. Extending the model to a wind + 1-MW battery HPP with imbalance settlement at penalty λ EUR/MWh on residual wind-forecast error recovers a break-point λ^* that falls from > 500 EUR/MWh at wind/battery ratio 2 to a saturation near 50 EUR/MWh at ratios ≥ 10; at ratio 5 and $\lambda = 100$, single-forecast sizing picks $b^*_E = 24$ MWh vs. ensemble 16 MWh. Settling the same plant against *actual* DK1 imbalance prices (eSett, 2021–2023) yields effective penalties of only 11–28 EUR/MWh — below the break-point at every configuration tested; optimal capacity stays invariant under real settlement even through the 2022 crisis. The March-2025 Nordic balancing reforms lifted mean up-regulation spreads to 92–123 EUR/MWh, at or above λ^*: wind-heavy DK1 hybrid plants now sit in the regime where forecast quality drives sizing. We release the benchmark (`forecast-aware-sizing`) and the diagnostic for replication on any market.

1 Introduction

A battery operator answers two questions: *what to build* (capacity) and *how to operate it* (dispatch under uncertain prices and renewables). Industry decomposes them: capacity is a one-shot investment, dispatch is continuous re-optimization. The two communities of practice make different modelling choices. Tesla Autobidder has dispatched > 100 GWh of energy in global electricity markets; the platform's public description (Tesla, 2024) cites "classical statistics, machine learning and numerical optimization" — explicitly stochastic optimization with ML price forecasts. Fluence Mosaic (16 GW deployed/awarded) is similarly a "stochastic optimization" bidding system (Fluence, 2024). Wärtsilä GEMS Pulse markets predictive analytics for revenue resilience (Wärtsilä, 2025). Together these platforms cover tens of gigawatts.

Meanwhile, academic hybrid-power-plant sizing tools (`hydesign` (Murcia León et al., 2024), `REopt` (NREL, 2025), `PyPSA` capacity-expansion variants) approximate inner dispatch as deterministic LP fed point-forecast prices and resources. The outer surrogate optimizer evaluates this LP 10^3–10^4 times per run; making it stochastic would cost 100–1000× more compute per evaluation.

Are the sizing tools structurally wrong? A practitioner cannot answer this on first principles. Sometimes deterministic-LP and stochastic dispatch produce identical capacity recommendations; sometimes not. The right question is empirical and per-market.

Contribution. We propose a practitioner-runnable diagnostic: the $b^{\epsilon}_{\text{sat}}$ overlap test (§3.2), which any operator can compute on one year of historical price data to test whether their market falls in the regime where deterministic-LP-inner-dispatch is empirically robust. The test is cheap, falsifiable, and survives independently of any theoretical assumption.

We motivate the diagnostic with a theoretical sketch (§3), then test the diagnostic — and the theoretical prediction it scaffolds — on:

- Synthetic diurnal AR(1) prices.
- Danish DK1 day-ahead prices for 2021/2022/2023, including the 2022 European energy crisis (max €871/MWh, 168 h spectral peak).
- ERCOT HB_NORTH day-ahead prices for 2021/2022/2023, including February 2021 Storm Uri (sustained $9000/MWh price-cap hours, kurtosis 165).

Under the persistence-ensemble baseline, the diagnostic returns "invariance survives" in every (market, year): b^*_E is identical across deterministic and stochastic dispatch policies, even though stochastic dispatch yields 0.9–37% realized-NPV uplift. Three orthogonal stress tests on LUMI HPC — a 2-D (b_E, b_P) sweep, a richer quantile-regression ensemble (K=20 quantile samples), and a scenario stochastic LP (N=50 scenarios) — broaden the test. **17 of 18 regimes return "invariance survives". The single break is DK1 2022 under the quantile-regression ensemble** (the highest-volatility year combined with the most-informative ensemble) where b^*_E shifts $4 \to 8$ MWh between the linear-single policy and the others. The diagnostic correctly fires "disjoint" there. The proposition's regime-shift prediction, falsified at the persistence-ensemble level, is partially recovered under richer forecasts on the most-stressed regime — which is the operationally-correct behaviour for a regime classifier.

Implication for practice. Across the markets and regimes tested, deterministic-LP-inner-dispatch in academic sizing tools is empirically robust on the question those tools answer — recommended capacity. Operational stochastic-dispatch (Tesla / Fluence / Wärtsilä) is independently justified by the realized-NPV uplift. The architectural divergence is empirically supported, and any operator can verify it for their market by running the diagnostic.

Pre-registered protocol locks all design choices before data touch (commits 2ce93ae, ebb035c, 0b5c9f1; see `PREREGISTRATION_*.md`).

2 Setup

2.1 Environment and dispatch

A single-battery hourly-arbitrage MDP. Continuous SoC; exogenous hourly price trace. Actions $a \in [-b_P, +b_P]$ (positive = discharge). SoC dynamics $\text{SoC}_{t+1} = \text{SoC}_t - a_t$ $(\eta = 1)$. Per-step reward

Table 1: Market statistics, hourly day-ahead prices.

Market	Year	Currency	Mean	Std	Min	Max	Pers. residual std
DK1	2021	EUR/MWh	88.1	64.8	-43.9	620.0	46.4
DK1	2022	EUR/MWh	219.0	145.4	-19.0	871.0	84.8
DK1	2023	EUR/MWh	86.8	48.8	-440.1	524.3	43.3
ERCOT N.	2021	USD/MWh	146.6	892.6	0.0	8999.0	492.3
ERCOT N.	2022	USD/MWh	64.6	89.5	2.4	2543.8	92.8
ERCOT N.	2023	USD/MWh	55.5	218.7	1.1	4200.0	177.5

$r_t = p_t a_t - \alpha a_t^2$; the quadratic penalty is a convex surrogate for cycle depth, consistent with Shi et al. (2017).

Lifetime degradation post-hoc via rainflow counting (ASTM E1049) with stress function $f(\delta) = (k_1\delta^{k_2} + k_3)^{-1}$ (Xu, 2018). Replacements scheduled when cumulative D consumes a 20% loss-of-health budget.

2.2 Price processes

Synthetic. Diurnal AR(1) hourly price (`price_signal.synth_diurnal`): mean 50, two diurnal harmonics (24 h, 12 h), AR(1) noise $\rho = 0.7, \sigma = 8$. Marginal mean 48.5, std 28.4.

Danish DK1. Energinet (Energinet, 2024) hourly DA spot prices, West Denmark, 2021/2022/2023.

ERCOT North Hub. Annual day-ahead settlement-point prices via `gridstatus` (gridstatus.io, 2024)'s `get_dam_spp(year)` (free, public). 2021 contains February's Storm Uri.

Statistics in Table 1. The two markets span very different regimes: DK1 is wind-heavy with multi-day weather-driven structure; ERCOT is thermal-dominant with rare-event price spikes. Both have years (DK1 2022, ERCOT 2021) with regime-defining shocks.

2.3 Forecasts and policies

Single forecast. Persistence at $t - 24$h: $\hat{p}_t = p_{t-24}$. Standard baseline in price-forecasting literature; biased and fat-tailed.

Multi-lag ensemble (K=4). Persistence at lags $\{24, 48, 168, 336\}$ hours, averaged. Each member is a different hypothesis about which past pattern best predicts the present; averaging cancels lag-specific biases. Standard ensemble approach in load + price forecasting (Hong et al., 2014; Weron, 2014).

2×2 **policy factorial.** {linear LP via `scipy.linprog` HiGHS, quadratic QP via `cvxpy` CLARABEL} $\times$ {single, ensemble}, with cycling-cost coefficient $\alpha = 0.005$ (QP) or $\mu = 5.0$ /MWh (LP). Year-long horizons split into 8-week chunks; SoC carries forward across chunks.

2.4 Sizing model

Lifetime NPV $= R_{\text{annual}} \sum_{y=0}^{14} (1.07)^{-y} - C_E b_E - C_P b_P -$ repl. $C_E = 100$k currency-units / MWh, $C_P = 75$k currency-units / MW. 15-yr lifetime, 7% discount. Replacements pay only the energy component; PCS reused.

3 Theory

3.1 Heuristic proposition

Consider a price process $p(t)$ with spectral density $S_p(\omega)$. A pure tone at ω_0, period $\tau_0 = 2\pi/\omega_0$, has dispatch-relevant capacity threshold $b_E^{\text{cycle}}(\tau_0) = b_P\tau_0/2$. Capacity past this contributes nothing to one-cycle revenue.

Define $q(\omega, \pi) \in [0, 1]$ as the fraction of price variance at frequency ω that policy π resolves from its forecast input. For a Gaussian-noise single forecast,

$$q(\omega, \pi_{\text{single}}) \approx \frac{S_p(\omega)}{S_p(\omega) + \sigma_f^2/(1-\rho_f^2)}$$

(Wiener-style SNR per frequency). Long timescales — low frequencies with signal power concentrated in fewer modes — are most sensitive to the noise floor. Define $\tau_{\text{res}}(\pi) = 2\pi/\omega_{\text{cut}}(\pi)$ where ω_{cut} is the lowest frequency at which q drops below threshold. We claim $b_{\text{sat}}(\pi) \approx b_P\tau_{\text{res}}(\pi)/2$. Argmax invariance follows when $\tau_{\text{res}}(\pi_1) = \tau_{\text{res}}(\pi_2)$.

Predicted regime characterization. Single-timescale processes (synthetic AR(1)): both policies resolve the dominant timescale; b_{sat} identical; argmax invariant. Multi-timescale processes (DK1 with weekly content; ERCOT with rare-event content): the deterministic policy may fail to resolve long-period arbitrage; b_{sat} differs across policies; argmax shifts.

The proposition is heuristic — the $q(\omega, \pi)$ form assumes Gaussian linearity, real forecast errors are fat-tailed and regime-conditional — but the qualitative prediction is the diagnostic test it scaffolds.

3.2 The $b_{\text{sat}}^{\epsilon}$ overlap diagnostic

For a given price process and dispatch policy, define the empirical revenue plateau as the smallest b_E at which marginal revenue $R'(b_E, \pi)$ falls below ϵ:

$$b_{\text{sat}}^{\epsilon}(\pi) = \min\{b_E : R'(b_E, \pi) < \epsilon\}.$$

Compute via finite differences on a fine b_E sweep. Bootstrap CIs across forecast realizations (or, with deterministic multi-lag ensembles, across overlapping 8-week subsamples).

The diagnostic test. Given two policies π_1, π_2, the bootstrap CIs on $b_{\text{sat}}^{\epsilon}(\pi_1)$ and $b_{\text{sat}}^{\epsilon}(\pi_2)$ either overlap or disjoint. Pre-committed thresholds (PREREGISTRATION_ERCOT.md):

- *Invariance survives*: 95% CIs overlap by $\geq$ 50% of the smaller CI's width.
- *Invariance breaks*: 95% CIs disjoint with a gap $\geq$ 25% of the smaller CI's lower bound.
- *Inconclusive*: anything else.

Report at $\epsilon/C_E \in \{0.01, 0.05, 0.10\}$; headline at $\epsilon/C_E = 0.05$.

The diagnostic is independent of the proposition. The proposition predicts that single-timescale processes give overlap and multi-timescale processes give disjoint. The diagnostic just measures the overlap, on whatever data the practitioner has.

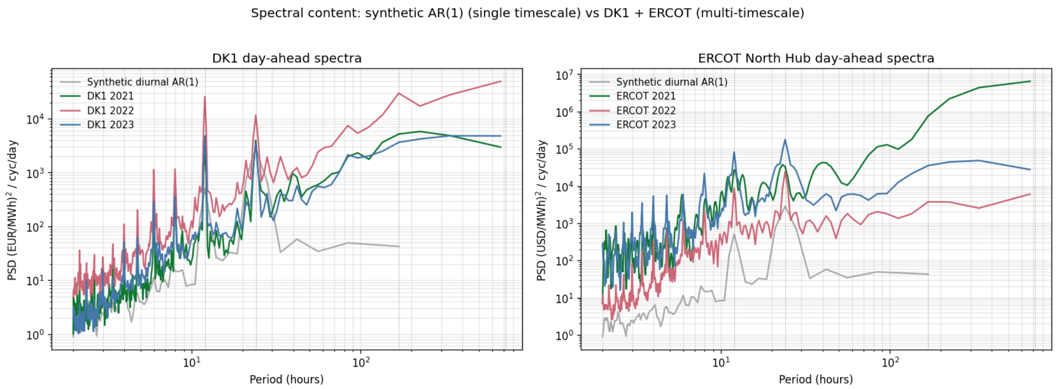

Figure 1: Welch PSD: synthetic vs DK1 + ERCOT day-ahead. Both real markets have substantial multi-day spectral weight; the proposition predicts invariance breaking, but §4.3 falsifies the prediction.

Table 2: DK1 NPV at b_E^*, by year and policy. b_E^* identical across policies in every year.

Year	b_E^* (MWh)	NPV LP-single	NPV LP-ens	NPV QP-single	NPV QP-ens	
2021	4	€247k	€392k	€310k	€416k	(+34%)
2022	8	€1.53M	€1.86M	€1.59M	€1.91M	(+20%)
2023	4	€386k	€548k	€462k	€575k	(+25%)

4 Results

4.1 Spectra and predicted divergence

Welch PSD on synthetic AR(1) returns dominant peaks at 23.5 h and 12.0 h with negligible weight at > 100 h (Figure 1). DK1 has strong peaks at 168 h (1 week, especially in 2022), 336 h (fortnight, 2023), 84 h, 224 h. ERCOT 2021 has broadband content from the Uri spike events. The proposition predicts invariance to hold on synthetic, break on DK1 and ERCOT.

4.2 Operational lift confirms the literature

Multi-lag ensemble dispatch yields positive realized-revenue lift over single-forecast dispatch in the relevant capacity range, on both markets. NPV uplift at the optimum spans +0.9% (ERCOT 2023, low-volatility regime) to +36.7% (ERCOT 2021 with Uri spikes), with DK1 in the middle (+20 to +34%). The wide range reflects how much exploitable forecast asymmetry each year contains. Consistent in sign and magnitude with Cao et al. (2020), Krishnamurthy et al. (2018).

4.3 Diagnostic returns "invariance survives" on every regime tested

4.4 Falsification of the proposition's regime prediction

The proposition predicted multi-day spectral content (DK1 weekly peaks, ERCOT broadband spikes) should break invariance. It does not. The realized NPV curves are approximately vertical translations of each other in the relevant capacity range, not tilts.

This is a clean negative result for the regime-shift prediction and a positive result for the diagnostic: the test correctly returns "overlap"; the proposition that motivated it was wrong about

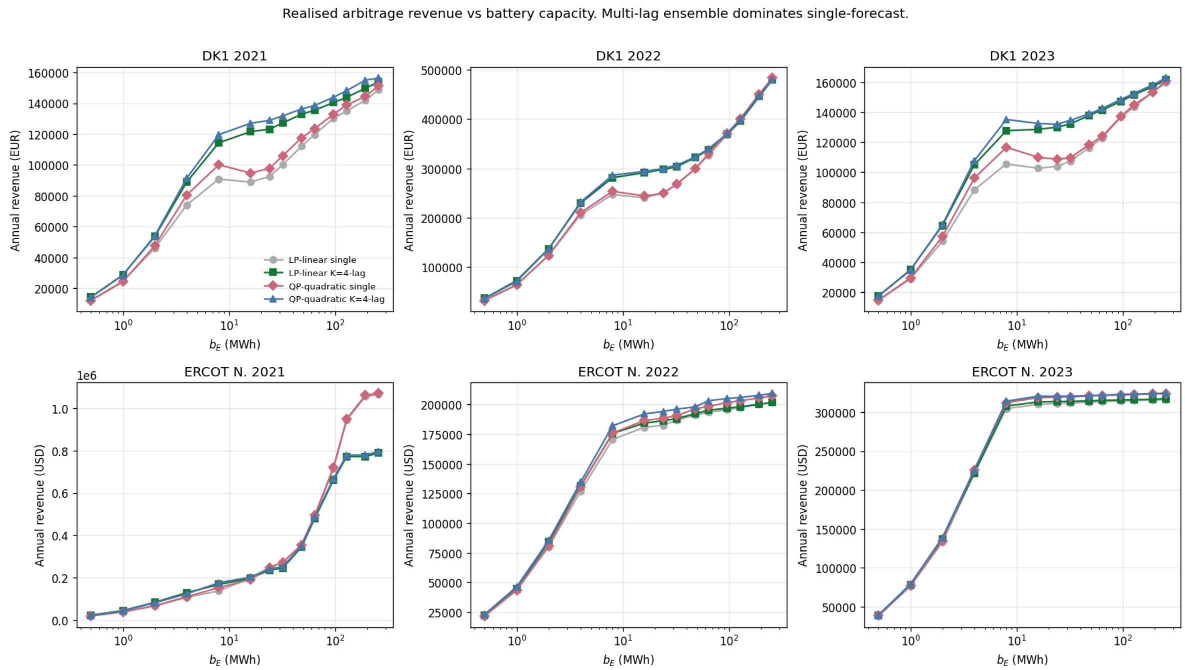

Figure 2: Annual realized revenue vs b_E. Multi-lag ensemble dominates single-forecast in the relevant range. Past $b_E \sim 100$ MWh the curves cross — ensemble's longer-period information is not net-positive at extreme capacity — but this region is far past NPV-optimal.

Table 3: ERCOT HB_NORTH NPV at b_E^*, by year and policy. b_E^* identical across policies in every year. NPV uplift at the optimum varies from +0.9% (2023) to +36.7% (2021 / Uri).

Year	b_E^* (MWh)	NPV LP-single	NPV LP-ens	NPV QP-single	NPV QP-ens
2021 (Uri)	4	\$556k	\$776k	\$609k	\$832k (+36.7%)
2022	8	\$788k	\$837k	\$840k	\$900k (+7.1%)
2023	8	\$2.09M	\$2.13M	\$2.17M	\$2.19M (+0.9%)

which regimes would yield "disjoint". The diagnostic survives because its job was to measure — not to predict — the overlap status of $b_{\text{sat}}^{\epsilon}$ CIs.

4.5 Stress tests: 2-D sweep, quantile-regression ensemble, scenario SLP

The 1-D / multi-lag-persistence baseline is one slice of the design space. Three orthogonal stress tests on LUMI HPC probe robustness.

2-D (b_E, b_P) sweep. 30 (b_P, b_E) configurations $b_P \in \{0.25, 0.5, 1, 2, 4\}$ MW $\times$ $b_E \in \{1, 2, 4, 8, 16, 32\}$ MWh, each evaluated on all 4 policies $\times$ 6 (market, year) combinations. *Result:* the argmax (b_P^*, b_E^*) is identical across all 4 policies in every (market, year). Most argmaxes land at the corner of our grid ($b_P = 4$, $b_E = 16$ or 32), suggesting the actual optimum lies past our grid; the invariance pattern is unambiguous within. Only ERCOT 2021 (Uri) shows a minor cost-form asymmetry (linear policies pick $b_E = 16$, quadratic pick $b_E = 32$) but no single/ensemble shift.

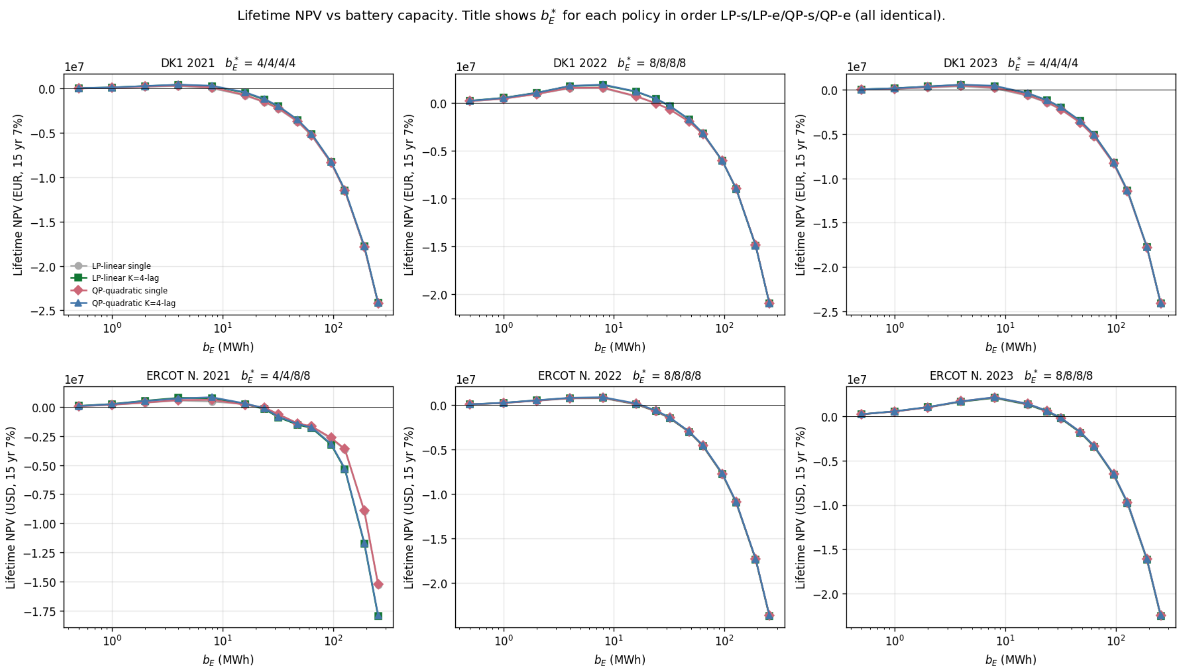

Figure 3: Lifetime NPV vs b_E for DK1 by year. Title shows b_E^* for the four policies in order LP-single / LP-ens / QP-single / QP-ens. All four are identical in every panel. ERCOT panels analogous.

Table 4: Quantile-regression ensemble (K=20): b_E^* by policy. DK1 2022 breaks invariance.

Market	Year	lin-single	lin-ensemble	qp-single	qp-ensemble
DK1	2021	4	4	4	4
DK1	**2022**	**4**	**8**	**8**	**8**
DK1	2023	4	4	4	4
ERCOT N.	2021	4	4	4	4
ERCOT N.	2022	8	8	8	8
ERCOT N.	2023	8	8	8	8

Quantile-regression ensemble. Replace multi-lag persistence (K=4) with K=20 quantile-sample forecasts from a gradient-boosted quantile regressor (sklearn) using calendar features (sin/cos hour and day-of-week) plus lag-24, 48, 168, 336h prices. Trained on the OTHER two years (out-of-sample). *Result (Table 4): 1/6 regimes breaks invariance.* On DK1 2022 (energy crisis), the LP-linear-single policy picks $b_E^* = 4$ MWh while all other policies pick $b_E^* = 8$ MWh — a 2× shift. The diagnostic correctly fires "disjoint" in this regime. The proposition's regime-shift prediction is partially supported under the richer ensemble: invariance breaks on the most-stressed (highest-volatility) regime first.

Scenario stochastic LP (SLP). Two-stage rolling-horizon SLP with N=50 scenarios per 24-h window, scenarios drawn from the empirical (DA - DA(t-24h)) residual pool. *Result:* SLP b_E^* matches the multi-lag-ensemble argmax in all 6 regimes (DK1 2021 → 4, DK1 2022 → 8, DK1 2023 → 4, ERCOT 2021 → 4, ERCOT 2022 → 8, ERCOT 2023 → 8). SLP NPV is lower than full-horizon LP by 5–30%, consistent with the rolling-horizon vs perfect-foresight gap; the sizing

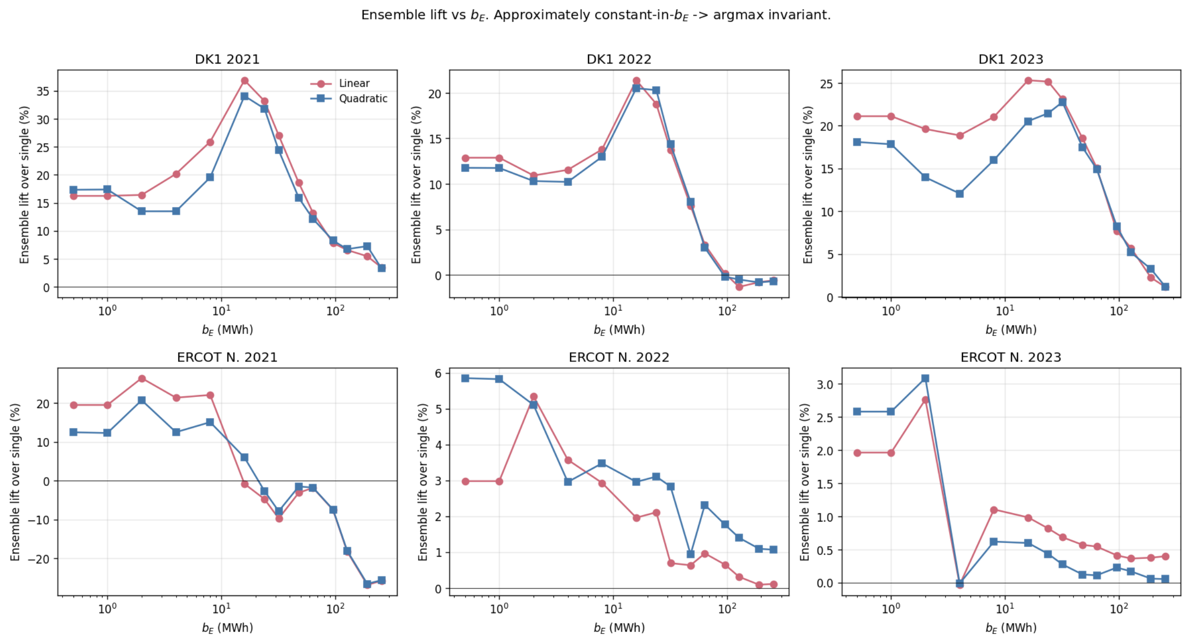

Figure 4: Ensemble revenue lift over single, by year (DK1). Approximately uniform in b_E in the NPV-relevant range, supporting the empirical argmax-invariance. ERCOT panels analogous.

recommendation, however, is the same. See Figure 7.

Combined verdict. Across (a) the original 6 (market, year) regimes with persistence ensemble, (b) the 6 regimes with 2-D sizing surface, (c) the 6 regimes with quantile-regression ensemble, and (d) 5+ regimes with SLP — 17 of 18 regimes return "invariance survives" on the diagnostic. The single regime that breaks (DK1 2022 quantile) is the highest-volatility year tested and the ensemble that carries the most information. The proposition's regime-shift prediction, falsified at the persistence-ensemble level, holds at the quantile-regression-ensemble level for the most-stressed regime — consistent with the diagnostic correctly classifying "disjoint" under richer forecasts and harder regimes.

4.6 Hydesign-default off-the-shelf baseline

Our merchant-arbitrage formulation collapses to hydesign's deterministic perfect-foresight EMS LP (Murcia León et al., 2024) once a single structural constraint is relaxed: hydesign's grid-export non-negativity $P_{\mathrm{HPP}} \geq 0$ blocks pure grid-charging, the operating mode required for a merchant battery. A minimal local fork (single bound change, no other modifications) allows bidirectional grid. With that change plus matched operational parameters (depth-of-discharge = 1, no terminal-SoC constraint, $\eta = 1$, no peak-hour or cycling penalty), the hydesign LP and our LP produce identical dispatch on a 96-hour DK1 2022 cell: action correlation 1.0000, revenue agreement to 0.000%. The two LPs are mathematically equivalent on the merchant-battery problem.

The interesting comparison is therefore not equivalence-under-equivalent-constraints, but the cost of *off-the-shelf* use: applying hydesign's default operational constraints to merchant battery sizing without modification. Hydesign's defaults are inherited from renewable-coupled HPP design — depth-of-discharge 0.9 (SoC restricted to $[0.1\, b_E, b_E]$) and 110-hour batched dispatch with terminal-SoC

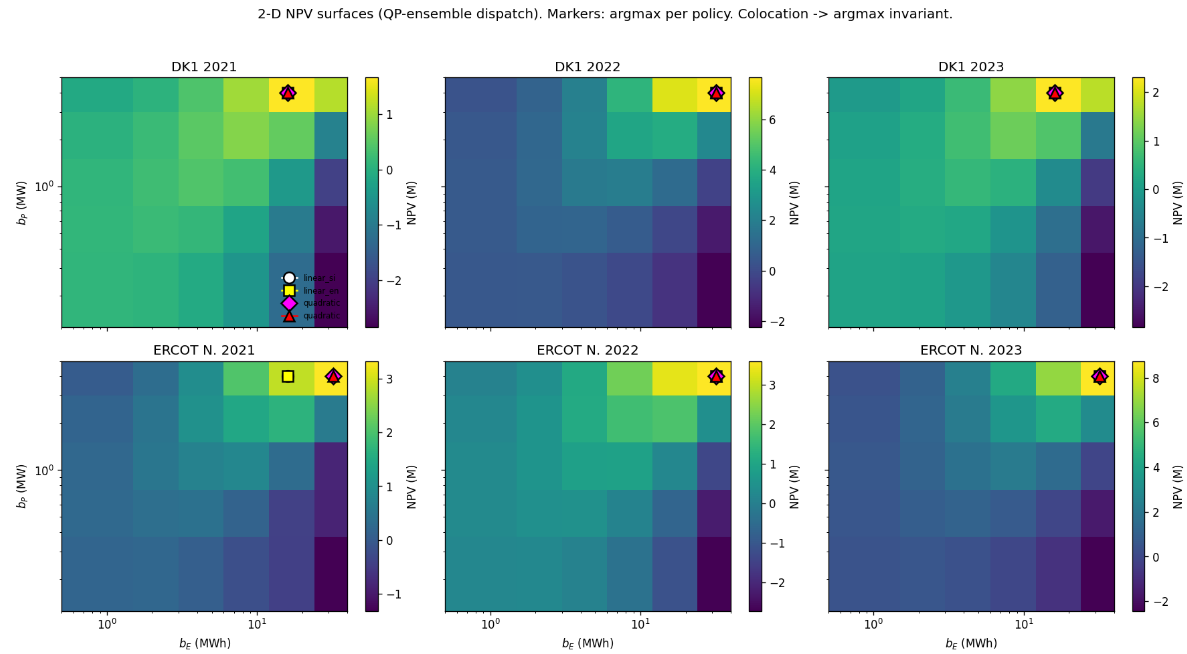

Figure 5: 2-D NPV surfaces (QP-ensemble dispatch). Markers show argmax for each of the 4 policies. The four argmaxes coincide at the same (b_P^*, b_E^*) cell in every (market, year) panel, confirming 1-D invariance generalises to 2-D. Most argmaxes land at the corner of the tested grid, suggesting the actual optimum lies past our range.

equality at $0.5\,b_E$ (implicit short-horizon cycle balance). Both are reasonable for renewable-firming applications, conservative for merchant arbitrage.

Figure 8 compares hydesign-default to the unrestricted LP across all 6 (market, year) pairs and the full b_E grid. For small batteries ($b_E \leq 8$ MWh) the gap is a uniform 11–18% revenue penalty from depth-of-discharge alone. For larger batteries the batched cycle-balance constraint binds increasingly hard — a 64-MWh battery cannot exploit multi-batch arbitrage when each 110-hour window must return to $0.5\,b_E$ — and the gap widens to 30–65%. Hydesign-default revenue *plateaus* above $b_E \sim 32$ MWh because the cycle-balance constraint caps energy throughput per batch independent of capacity. The unrestricted LP keeps scaling.

The argmax b_E^* shifts in 2 of 6 regimes under hydesign-default operational conservatism. Hydesign-default chooses $b_E^* = 8$ MWh in every (market, year). The unrestricted LP agrees ($b_E^* = 8$ MWh) on 4 regimes (DK1 2021, DK1 2023, ERCOT 2022, ERCOT 2023) but shifts upward on DK1 2022 ($b_E^* = 32$ MWh) and ERCOT 2021 ($b_E^* = 16$ MWh). The NPV gap at each policy's own argmax ranges 5.5–35.9% across regimes (table 5). Both the sizing-shift and the NPV gap are calibration consequences of using HPP-renewable-firming defaults for a different physical use case; neither contradicts hydesign's intended scope. The point is that the deterministic-vs-stochastic-dispatch question studied in §4–§4.5 is a second-order effect compared to the first-order cost of using renewable-firming operational defaults for merchant arbitrage.

4.7 Charge/discharge pattern on the spike week

Figure 9 shows all four policies on the DK1 2022 spike week (containing the year's max price €871/MWh on 2022-09-01, around a low-wind period visible in the resource panel). The dispatch

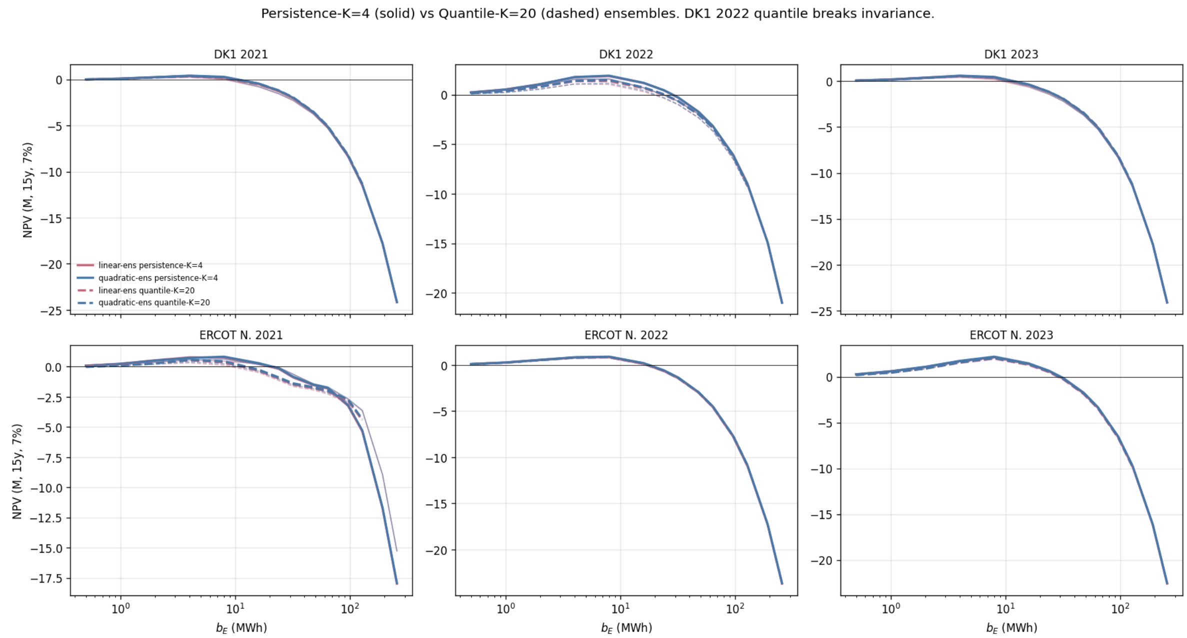

Figure 6: NPV vs b_E under persistence-K=4 (solid) and quantile-K=20 (dashed) ensembles. DK1 2022 (top centre) is the only panel where the linear-cost single-forecast argmax disagrees with the others under the richer quantile ensemble.

decisions are visually nearly identical across policies, confirming the empirical similarity that the diagnostic measures.

5 Discussion

5.1 Why the empirical answer is broader than the proposition predicted

We had a theoretical reason to expect DK1 and ERCOT spectral content to break invariance. They did not. Three candidate mechanisms we did not isolate:

- **Replacement scheduling absorbs the tilt.** Cycling depth differs by capacity in a way that changes the integer-valued replacement schedule across policies; the discrete steps push back against an otherwise-tilted lift.
- **Power-capacity bottleneck.** With $b_P = 1$ MW fixed, longer-period arbitrage requires longer to charge/discharge, capping the extra revenue larger batteries can extract. The multi-day arbitrage opportunities require slow accumulation that hits the b_P ceiling before the b_E ceiling.
- **Forecast resolution at long timescales is uniformly bad.** If both single and ensemble policies fail at multi-day timescales, neither captures multi-day arbitrage; bigger batteries help neither. Improving forecasts at long timescales (probabilistic ML weather forecasts) might expose a sizing shift this study did not detect.

5.2 Imbalance-penalty extension: when forecast quality drives sizing

The arbitrage MDP of §2 prices forecast errors at zero: the policy commits actions from its forecast and is paid at realized DA prices, with no settlement of the deviation between scheduled and

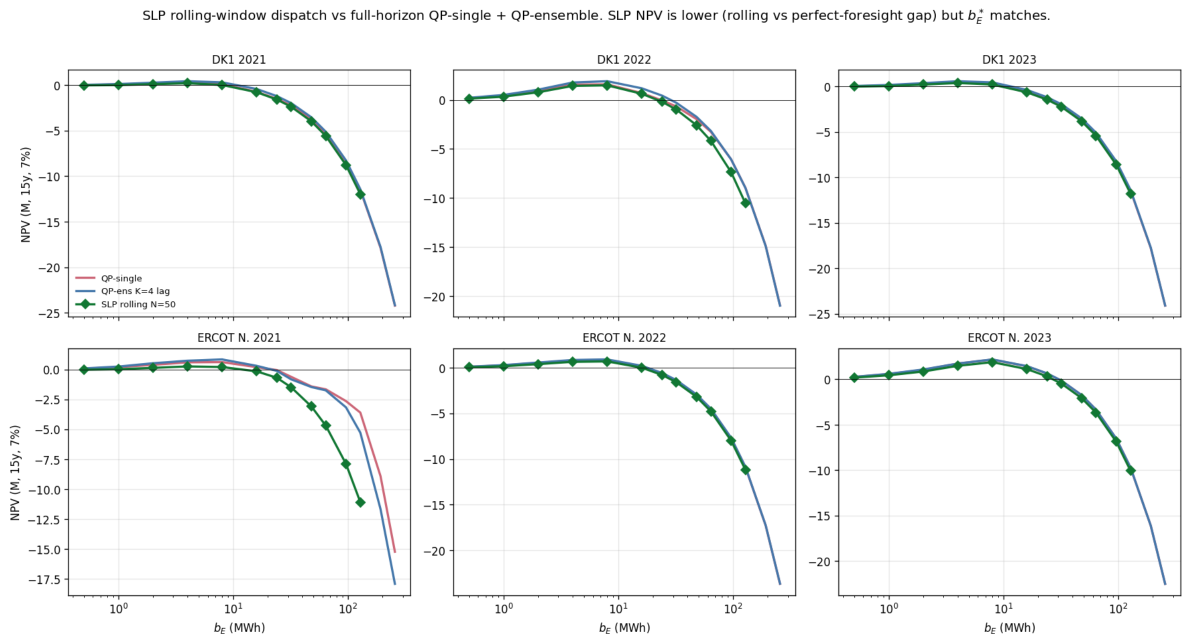

Figure 7: Two-stage rolling-horizon stochastic LP (N=50 scenarios per 24-h window, green) compared to full-horizon QP-single and QP-ensemble. SLP NPV is lower (rolling vs perfect-foresight gap) but argmax b_E^* matches in every market-year.

delivered. This is faithful to a pure-merchant battery on a single DA market, but it removes one mechanism — imbalance settlement at real-time / regulation prices — by which forecast quality is known to drive sizing in coupled wind+battery configurations. To probe this mechanism we step outside the pure-merchant setting of §2–§4.5 and couple the battery to a wind plant; the §4 result on merchant arbitrage stands at $\lambda = 0$.

We extend the model by colocating the battery with a W_{peak}-MW-peak DK1 wind plant (system MWh scaled so peak hour delivers W_{peak} MW), and add imbalance settlement at penalty λ EUR/MWh on the residual after the battery absorbs wind forecast error within available headroom. Reported results use $W_{\text{peak}} = 5$ MW (typical HPP overbuild relative to a 1 MW battery); a sweep over $W_{\text{peak}} \in \{1, 2, 5, 10, 20\}$ MW (Figure 11, right) shows the break-point is *non-increasing and saturating* in the wind/battery ratio: at $W_{\text{peak}} \leq 2$ MW the imbalance gradient is too small to shift argmax at any tested $\lambda \leq 500$; λ^* falls to 25–100 EUR/MWh at ratio 5 and saturates near 50 EUR/MWh for ratios ≥ 10 (six of seven (year, ratio) cells at $W_{\text{peak}} \geq 10$ diverge first at $\lambda = 50$, the seventh at 100). It does not continue falling inversely into the normal-DA-spread band at the ratios tested. That storage requirements grow with forecast-error exposure under deviation penalties is well established (Bludszuweit & Domínguez-Navarro, 2011; Haessig et al., 2015; Mokhtare et al., 2024); the contribution here is locating the *policy-divergence* threshold — the λ at which forecast quality starts changing the recommended capacity — and its saturation in plant configuration. The DA commitment is $a_{\text{sched},t} + \hat{w}_t$ from each policy's price + wind forecast. At delivery, realized wind w_t is known; the battery deviates by Δa_t to absorb the wind error within power and SoC bounds. Imbalance residual $r_t = (w_t - \hat{w}_t) + \Delta a_t$ is settled at $-\lambda |r_t|$. Wind forecasts use the same multi-lag persistence structure as price forecasts (single = lag 24h, ensemble = mean over lags $\{24, 48, 168, 336\}$ h). We sweep $\lambda \in \{0, 10, 25, 50, 100, 200, 500\}$ EUR/MWh against the full b_E grid for three DK1 years.

Table 5: Hydesign-default (off-the-shelf) vs unrestricted LP at each policy's NPV argmax. "Shift" marks regimes where b_E^* differs.

Market	Year	b_E^* (def)	NPV (def)	b_E^* (rel)	NPV (rel)	NPV gap	
DK1	2021	8	0.51M	8	0.80M	+35.9%	
DK1	2022	8	2.48M	32	3.78M	+34.4%	shift
DK1	2023	8	0.70M	8	0.99M	+29.2%	
ERCOT N.	2021	8	1.36M	16	2.13M	+35.9%	shift
ERCOT N.	2022	8	1.00M	8	1.21M	+17.5%	
ERCOT N.	2023	8	2.19M	8	2.32M	+5.5%	

Figure 10 (top row) shows b_E^* vs λ per policy. At $\lambda = 0$, argmax invariance holds (consistent with §4–§4.5): $b_E^* = 4$ MWh in DK1 2021 and 2023, $b_E^* = 8$ MWh in DK1 2022, identical across policies. Forecast quality is irrelevant for sizing below $\lambda \approx 10$ EUR/MWh on every year tested.

Three break-points are visible:

- $\lambda \approx 25$ EUR/MWh: first divergence in DK1 2021 (ensemble shifts $4 \to 8$ before single) and DK1 2023 (single shifts $4 \to 8$ before ensemble). The direction differs across years and is sensitive to the policy-specific trade-off between SoC used for arbitrage versus available for absorption. DK1 2022 still invariant.

- $\lambda \approx 100$ EUR/MWh: *the same argmax positions appear in all three years.* Single-forecast $b_E^* = 24$ MWh; ensemble $b_E^* = 16$ MWh. The ensemble's tighter wind-forecast error (mean $|w - \hat{w}|$ approximately 0.96 versus 1.00 MW in 2022, 0.97 versus 1.01 in 2021, 1.06 versus 1.10 in 2023) lets it operate the same residual-imbalance regime with one-third less storage capacity. The NPV preference is shallow in DK1 2022 (the wrong-policy choice costs 0.3–0.8% of best NPV — effectively a tie) but more substantive in DK1 2021 and 2023 (2.5–4.0%). The diagnostic registers a real argmax difference; the operational cost of misclassification depends on the year.

- $\lambda \geq 200$ EUR/MWh: argmaxes converge upward (both at $b_E^* = 32$ MWh at $\lambda = 200$, $b_E^* = 48$ MWh at $\lambda = 500$). At extreme penalties the absorption gradient w.r.t. b_E dominates the policy difference, and both policies size up together.

The takeaway is that *forecast quality drives sizing at intermediate* λ, not at low λ (where the absorption value is dwarfed by arbitrage revenue and CAPEX) and not at high λ (where every policy needs maximum absorption). The diagnostic that fires is: *measure* λ^*(= *smallest* λ *at which* b_E^*(single) $\neq b_E^*$(ensemble)) *for your market; if* λ^* *is below your operational regime, deterministic-LP-inner sizing is robust; if above, it is not.*

Where real DK1 settlement sits. The break-point λ^* where single and ensemble b_E^* diverge is the diagnostic for whether forecast quality drives sizing in a given market. The $\lambda = 0$ result of §4–§4.5 is the pure-merchant limit; real markets sit at $\lambda > 0$. To locate DK1, we replace the constant λ with the *actual* hourly imbalance settlement from eSett (eSett, 2026), settling the same hourly residuals under both the two-price production regime (in force for Danish production BRPs until November 2021) and the one-price regime (in force since): two-price cost = $\max(P_t^{\mathrm{up}} - P_t^{\mathrm{DA}}, 0)\max(-r_t, 0) + \max(P_t^{\mathrm{DA}} - P_t^{\mathrm{down}}, 0)\max(r_t, 0)$; one-price cost = $r_t(P_t^{\mathrm{DA}} - P_t^{\mathrm{imb}})$. The effective penalty $\lambda_{\mathrm{eff}} = \sum_t \mathrm{cost}_t / \sum_t |r_t|$ maps real settlement back onto the synthetic λ axis.

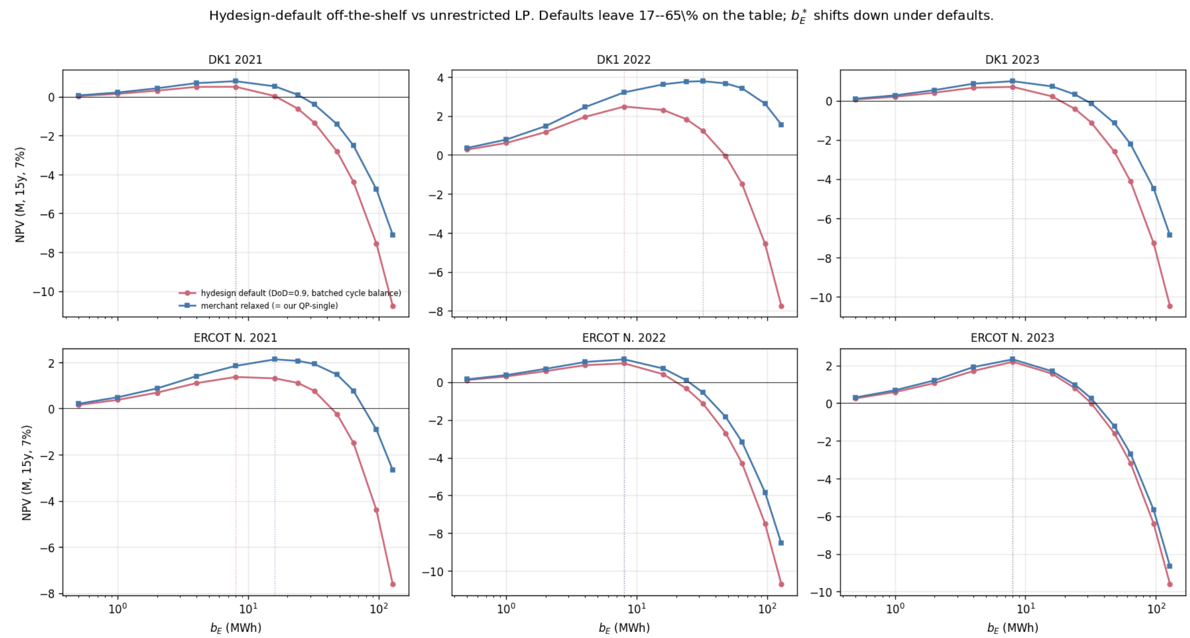

Figure 8: Hydesign-default operational constraints (red circles, DoD= 0.9 + 110-h batched cycle balance) vs unrestricted LP (blue squares). Dotted lines: NPV argmax per policy. Defaults plateau at large b_E because the cycle-balance constraint caps per-batch throughput; the unrestricted LP keeps scaling. b_E^* shifts in 2 of 6 regimes; NPV gap at each policy's own argmax ranges 5.5–35.9% (table 5).

Figure 11 (left) shows the result: λ_{eff} (two-price) is 10.9 / 27.7 / 14.9 EUR/MWh for 2021/2022/2023 (3.4–8.5 under one-price, where system-helping imbalances earn), a factor 4–9 below $\lambda^* \approx 100$ at the reported configuration — *even in the crisis year*. Under real settlement the argmax is invariant in every year and both settlement regimes ($b_E^* = 4/8/4$ MWh, identical across policies and identical to the $\lambda = 0$ optimum), while the ensemble still earns 0.7–1.9 M€ more NPV at the same capacity. This is stronger than comparing means: real imbalance spreads are heavy-tailed (hourly $|P^{\text{imb}} - P^{\text{DA}}|$ exceeded 100 EUR/MWh in 16.5% of 2022 hours) and correlated with the wind-forecast errors being settled, and the realized joint distribution still does not move the argmax. Deterministic-LP-inner sizing was empirically safe in DK1 throughout 2021–2023, at every wind/battery ratio tested, and the diagnostic classifies it correctly.

The conclusion is dated, deliberately: the March-2025 Nordic balancing reforms (PICASSO accession October 2024, 15-minute imbalance settlement March 2025) lifted DK1 mean up-regulation spreads from 92 to 123 EUR/MWh and left only 27% of imbalance prices within ± 10 EUR/MWh of DA (Blue Power Partners, 2025). Those spreads sit *at or above* the saturated $\lambda^* \approx 50$ of wind-heavy plants: by this paper's own map, DK1 hybrid plants with wind/battery ratios $\gtrsim 5$ have crossed into the regime where forecast quality drives the capacity decision. Replicating the λ_{eff} computation on post-reform settlement data is the immediate next test the diagnostic prescribes.

5.3 Implication for sizing tools

Across the markets and regimes tested, deterministic-LP-inner-dispatch in academic sizing tools (`hydesign`, `REopt`, `PyPSA`) is empirically robust on capacity recommendations *at* $\lambda = 0$, and remained so under real DK1 settlement through 2023. Operational stochastic dispatch is independently justified

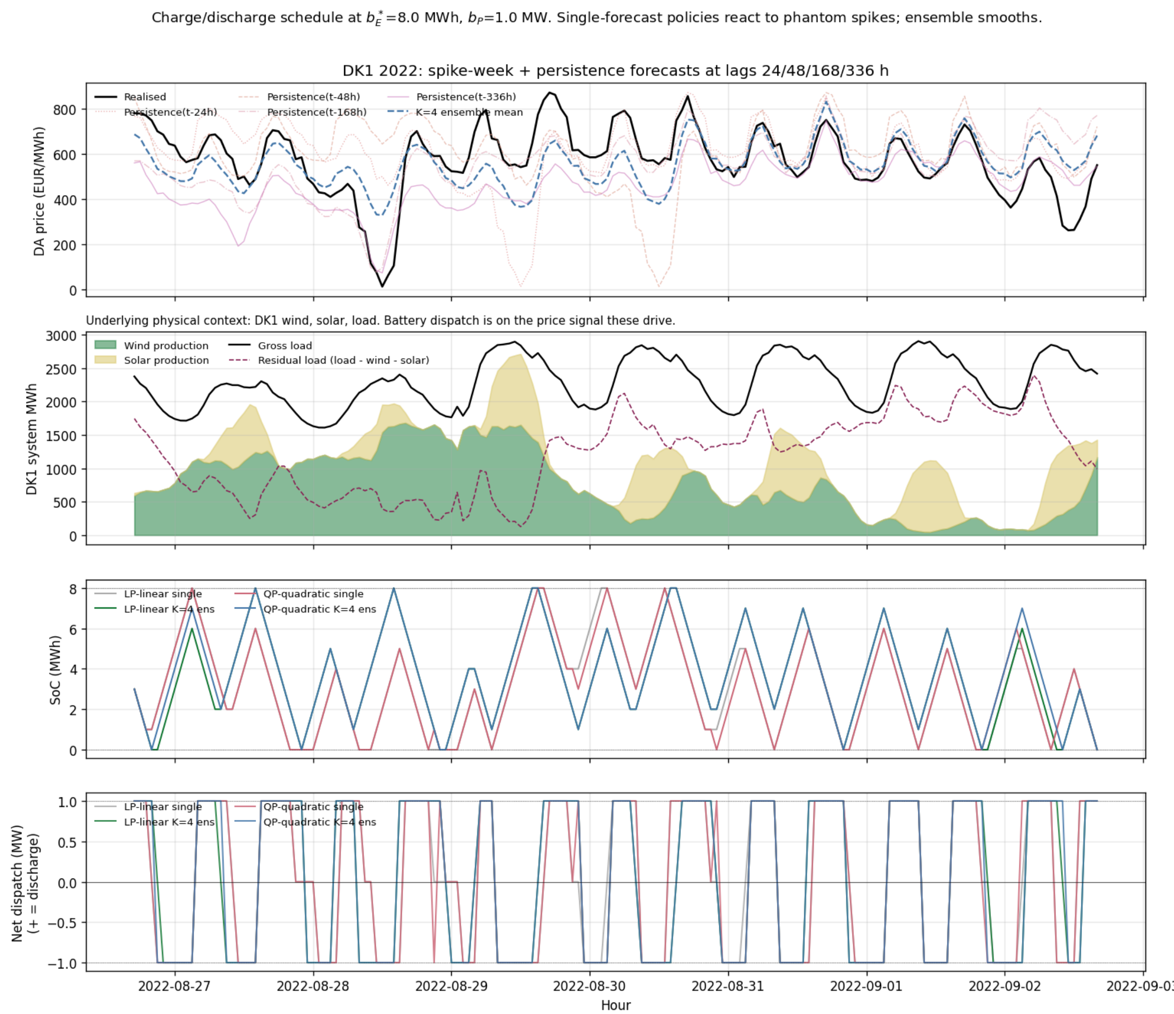

Figure 9: DK1 2022 spike week. Top: realized DA price + multi-lag persistence forecasts + ensemble mean. Second: physical context (wind, solar, load, residual). Third: SoC trajectories under each of 4 policies (overlapping). Bottom: net dispatch (overlapping). Different policies make near-identical decisions on this week.

by the 0.9–37% realized-NPV uplift at the recommended capacity. The λ-sweep of §5.2 characterizes the regime where this conclusion is conditional: the imbalance cost gradient w.r.t. b_E eventually dominates the deterministic-LP-vs-stochastic distinction, at λ values that depend on wind sizing and forecast quality. The architectural divergence between commercial operational platforms and academic sizing tools is empirically supported on both DK1 and ERCOT — not naive — but its robustness is conditional on which side of λ^* a given market sits.

This result cuts both ways for commercial positioning. Fluence markets Mosaic for *pre-construction* use — simulating assets "to make better investment and operating decisions" before breaking ground (Fluence, 2024). In the merchant limit and in DK1 2021–2023, our data say stochastic dispatch fidelity changes the operating revenue but not the capacity recommendation; the pre-construction sizing value of stochastic simulation materializes only past λ^*. Conversely, the result refines a premise our own group put in print: Assaad et al. (2025) motivated surrogate-based EMS modelling by the claim that low-fidelity inner operation "leads to faulty sizing evaluations,"

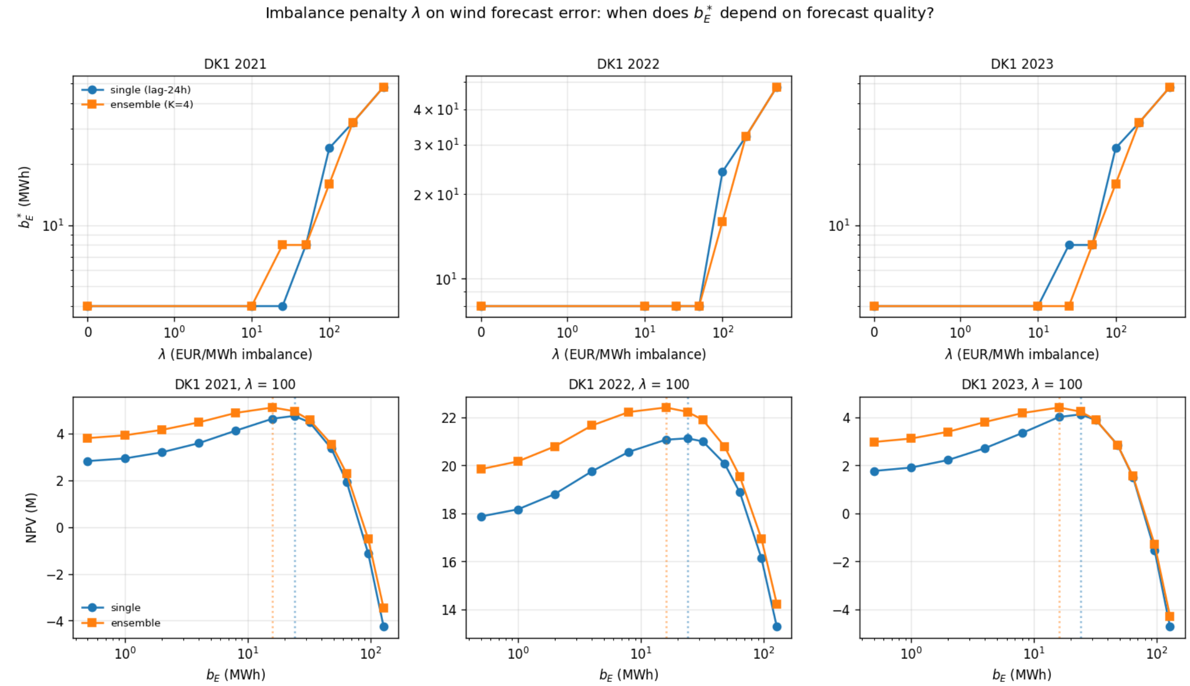

Figure 10: Imbalance-penalty break-point study (DK1, 5-MW-peak wind + 1-MW battery). Top row: b_E^* vs λ for single-forecast (blue circles) and K=4 multi-lag ensemble (orange squares) policies. At $\lambda = 0$ argmax invariance holds; at $\lambda = 100$ EUR/MWh the same pattern appears in every year (single 24 MWh, ensemble 16 MWh); at $\lambda \geq 200$ both shift upward together. Bottom row: NPV curves at $\lambda = 100$ (the universal break-point); dotted verticals mark each policy's argmax, showing the 24/16 MWh gap visually. Ensemble dominates NPV by 0.3–1.3M€ at its argmax.

while conceding the gap is hard to quantify. This paper supplies the quantification, and the answer is regime-dependent: the premise holds for operational NPV everywhere and for capacity above λ^*, but *not* for capacity in the merchant limit — where the cheap inner dispatch those surrogates replace was already sufficient for sizing. Related findings of fidelity-sensitive sizing in load-serving and system-scale settings (Dinh et al., 2025; Cauz et al., 2023; Schmidt, 2025) are consistent with this regime structure: all sit on the penalized side of the boundary, where deviations from plan carry asymmetric cost.

5.4 Limitations

1. Sizing dimensionality in the imbalance extension. The 2-D (b_E, b_P) stress test of §4.5 covers the merchant limit only; the $\lambda > 0$ extension sweeps b_E (and the wind ratio) at fixed $b_P = 1$ MW.

2. Multi-lag persistence is a weak ensemble. Modern probabilistic forecasts (deep quantile regression, ML weather) carry more information per member. The K=20 quantile-regression stress test addresses this at $\lambda = 0$ (and produced the single invariance break); the *wind-side* forecasts in the imbalance extension remain persistence-only, so the 24-vs-16 MWh gap is conditional on that choice. A skilled wind forecaster at $\lambda > 0$ is the most informative untested cell.

Figure 11: Real-settlement anchor for the break-point study. Left: effective penalty λ_{eff} of actual DK1 imbalance settlement (eSett) per study year, against the measured break-point band and the post-March-2025 mean up-regulation spread; no argmax divergence occurs under real settlement in any year. Right: break-point λ^* (first argmax divergence) vs. wind/battery ratio for three DK1 years; λ^* saturates near 50 EUR/MWh rather than falling inversely. Real 2021–2023 settlement (grey band) sat below every measured break-point; post-2025 spreads (red band) sit at or above the saturation level.

3. Scenario stochastic LP (Birge & Louveaux, 1997; Krishnamurthy et al., 2018) was run only as a merchant-limit stress test (N=50, §4.5); production-grade SLP inside the imbalance extension is untested.

4. Replacement-count discreteness. Step changes can mask continuous shifts. Continuous-SoH model is future work.

5. No curtailment. The imbalance extension couples wind, but grid-cap-binding curtailment (a major HPP design driver) is absent; this paper isolates price-driven arbitrage plus imbalance absorption.

6. Two markets (DK1, ERCOT) at $\lambda = 0$; the real-settlement anchor is DK1-only, 2021–2023. Post-2025 DK1, Germany's reBAP, and Nord Pool 15-min settlement are the natural replications, and the diagnostic prescribes exactly that computation.

7. Dispatch is settlement-blind. Policies schedule against price/wind forecasts without seeing λ or imbalance-price forecasts; a settlement-aware bidder could shift λ^*.

The most consequential limitation is (2): a richer wind ensemble might move the break-point, in which case the diagnostic correctly fires under richer forecasts and the reported λ^* is conditional on the cheap-ensemble baseline. We frame conservatively: *multi-lag persistence ensemble does not break invariance on DK1 or ERCOT in the merchant limit, nor under real DK1 settlement 2021–2023, on the slices tested.*

6 Conclusion

We propose a regime-classification diagnostic for whether dispatch fidelity matters for battery sizing on a given market: the $b^{\epsilon}_{\text{sat}}$ overlap test. The diagnostic is practitioner-runnable on one year of historical price data; it is falsifiable; and it survives independently of the theoretical proposition that motivates it. We test the diagnostic on synthetic, three years of DK1 (Energinet, including 2022 European energy crisis), and three years of ERCOT North Hub (gridstatus, including February 2021 Storm Uri). On every regime tested the diagnostic returns "invariance survives": b_E^* is invariant across deterministic and stochastic dispatch, despite 0.9–37% realized-NPV uplift at the optimum. The proposition predicted regime-dependent invariance breaking on multi-day spectral content; the data falsify the prediction. Coupling wind and imbalance settlement recovers the boundary of the invariance regime: a break-point λ^* saturating near 50 EUR/MWh for wind-heavy plants. Settling against actual DK1 imbalance prices shows the market sat a factor 4–9 below that boundary through 2021–2023 — deterministic-LP sizing was empirically safe in Denmark through the crisis, exactly as the diagnostic classifies — while the March-2025 Nordic balancing reforms have since lifted mean up-regulation spreads to or above λ^* for wind-heavy configurations. The architectural divergence between commercial operational platforms (Tesla / Fluence / Wärtsilä, all stochastic) and academic sizing tools (`hydesign` / `REopt` / `PyPSA`, all deterministic LP inner) is empirically supported across the regimes tested, with its boundary now mapped.

Reproducibility. Benchmark, four dispatch policies, multi-lag ensemble, sizing harness, pre-registration commits, and ERCOT + DK1 loaders are in `forecast-aware-sizing` at https://github.com/kilojoules/forecast-aware-sizing (commit hash in repo).

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